6 Rigging


Rigging Characters

Deformations change the shape of an object. Almost any sort of organic animation from a fully skinned character to flowers swaying in the breeze will need deformations of one sort or another. In addition to using deformations for animation tasks, you can use them as modelling tools. Because deformations can reshape a lot of detail quickly, deformers are good choices for global changes to an object. Rigging is primarily used in character animation to create hierarchical structures called skeletons. Skeletons are used as a framework for deforming the character as well as animating it. A good rig builds upon the skeleton to provide additional tools that make the animator's job easier by allowing the character to be quickly posed and manipulated.

6.1 Bones

Skeletons in 3ds Max can be constructed from any object but are usually constructed from objects called bones, which are tied together in a hierarchy. The skeleton, in turn, is used to deform a mesh via 3ds Max's skinning tools. Although skeletons are used primarily for character animation, they can also be used to create all sorts of other deformations. A garden hose, for example, can easily be deformed by using a series of bones. Bones can also help refine the behaviour of hair and clothing. Bones are used to guide deformations when using skinning tools. When the bones move, the mesh deforms to match. Other objects besides bones (such as boxes or other bits of geometry) can be used in skeletons. When building a skeleton, it is always a good idea to study the anatomy of the character or creature you are rigging. Getting the skeleton anatomically correct will make the resulting deformations anatomically correct as well. The bones of this character closely mimic the bones of a real skeleton.

6.1.1 Creating Bones

To create bones, go to Create > Systems > Bones IK Chain menu or under